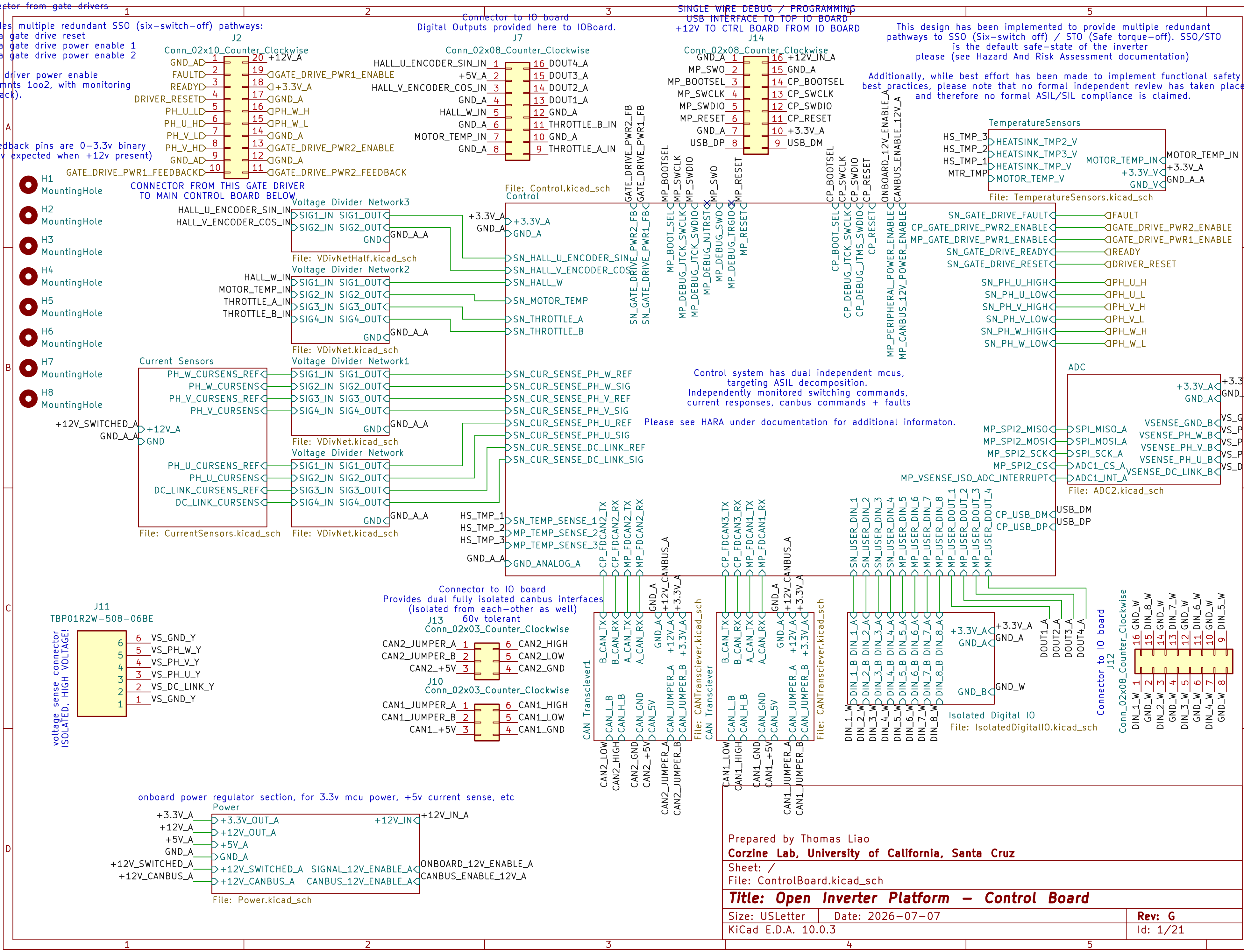
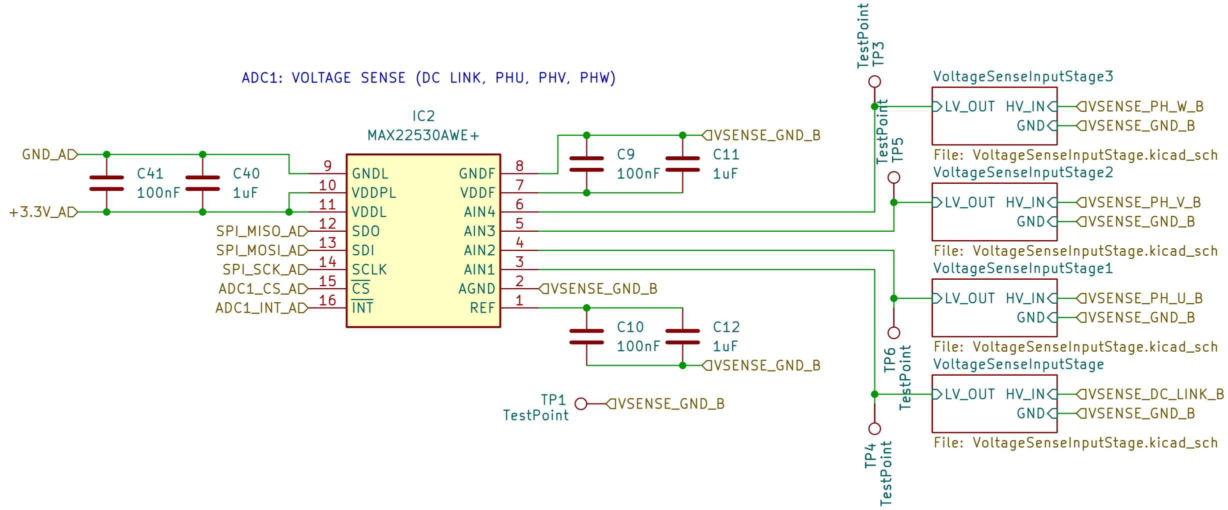
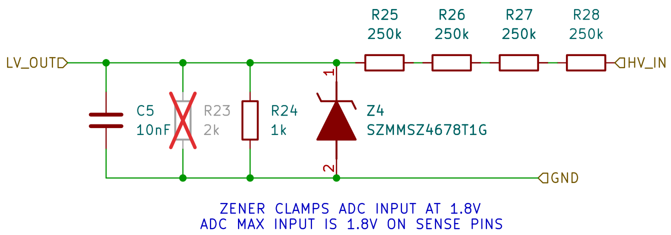


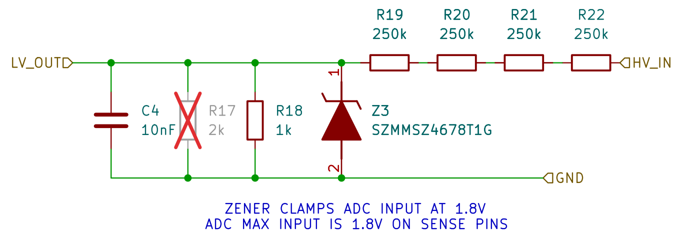
Connector from gate drivers
Includes multiple redundant SSO (six-switch-off) pathways:
- via gate drive reset
- via gate drive power enable 1
- via gate drive power enable 2
(Gate driver power enable implemnts 1oo2, with monitoring feedback).
feedback pins are 0-3.3v binary (3.3v expected when +12v present)
CONNECTOR FROM THIS GATE DRIVER TO MAIN CONTROL BOARD BELOW
Connector to IO board
Digital Outputs provided here to IOBoard.
SINGLE WIRE DEBUG / PROGRAMMING
USB INTERFACE TO TOP IO BOARD
+12V TO CTRL BOARD FROM IO BOARD
This design has been implemented to provide multiple redundant pathways to SSO (Six-switch off) / STO (Safe torque-off). SSO/STO is the default safe-state of the inverter please (see Hazard And Risk Assessment documentation)
Additionally, while best effort has been made to implement functional safety best practices, please note that no formal independent review has taken place, and therefore no formal ASIL/SIL compliance is claimed.
Control system has dual independent mcus, targeting ASIL decomposition. Independently monitored switching commands, current responses, canbus commands + faults
Please see HARA under documentation for additional informaton.
Connector to IO board
Provides dual fully isolated canbus interfaces (isolated from each-other as well)
60v tolerant
voltage sense connector
ISOLATED, HIGH VOLTAGE!
Connector to IO board
onboard power regulator section, for 3.3v mcu power, +5v current sense, etc
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: ControlBoard.kicad_sch
Title: Open Inverter Platform - Control Board
Size: USLetter
Date: 2026-07-07
Rev: G
KiCad E.D.A. 10.0.3
Id: 1/21



Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/
File: ADC2.kicad_sch
Title: Open Inverter Platform – Control Board
Size: USLetter | Date: 2026-07-07 | **Rev: G**
KiCad E.D.A. 10.0.3 | Id: 4/21



Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage/
File: VoltageSenseInputStage.kicad_sch
Title: Open Inverter Platform - Control Board
Size: USLetter | Date: 2026-07-07 | **Rev: G**
KiCad E.D.A. 10.0.3 | Id: 2/21

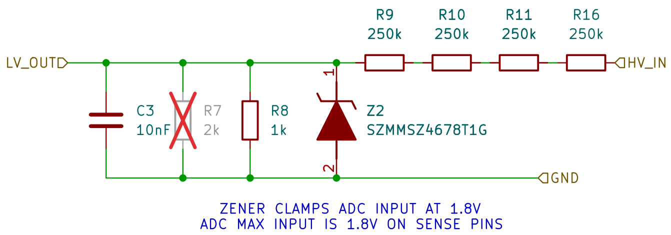


Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage1/
File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform — Control Board

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 3/21

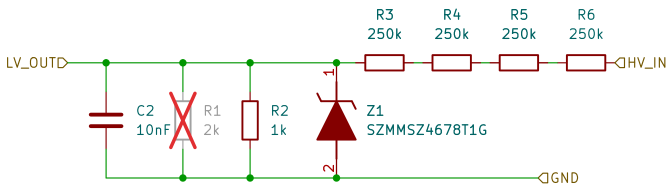


Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage2/
File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter | Date: 2026-07-07 | **Rev: G**
KiCad E.D.A. 10.0.3 | Id: 5/21



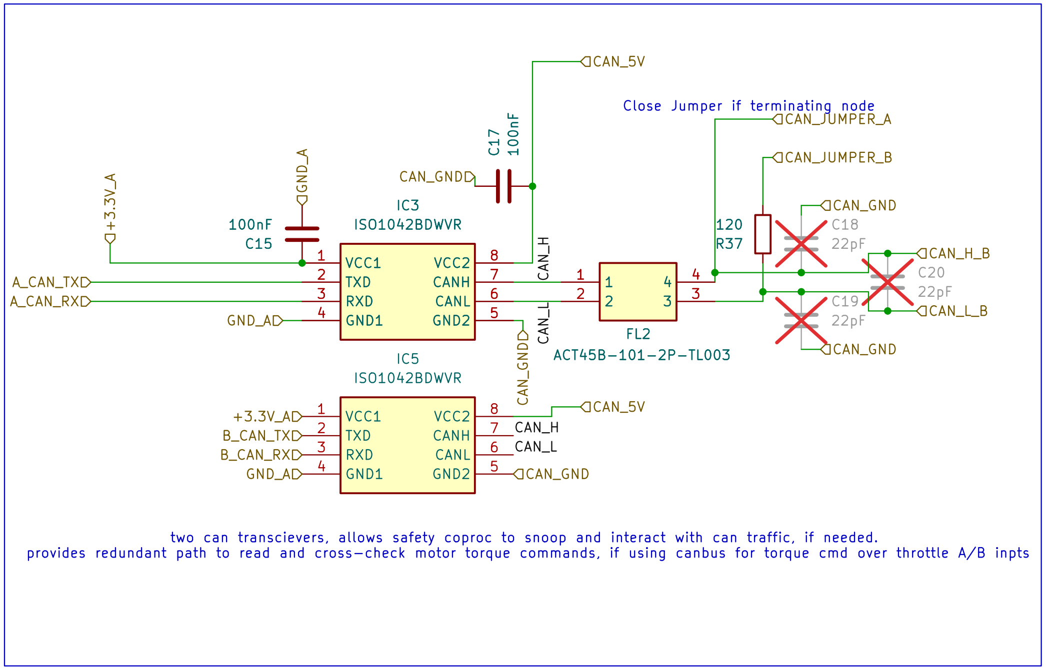
ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

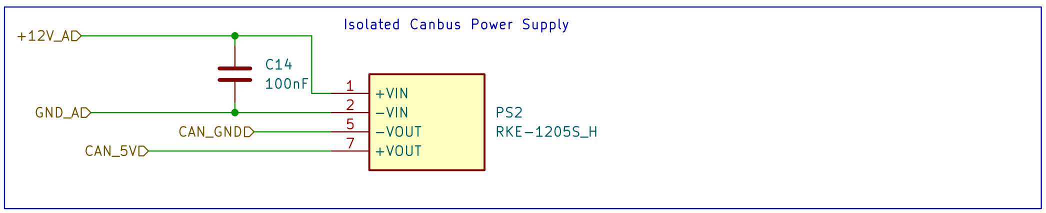
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage3/
File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 6/21





Prepared by Thomas Liao

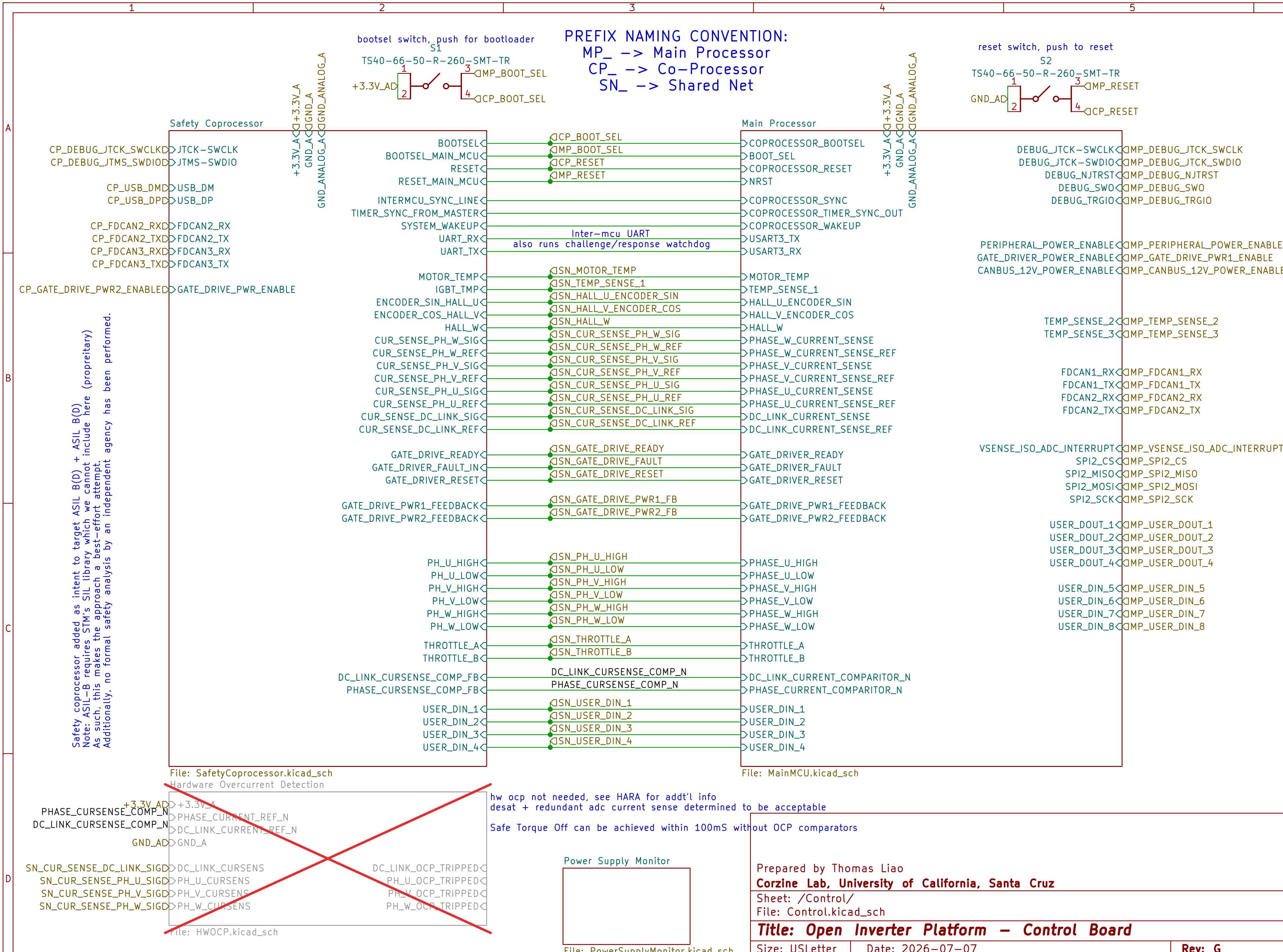
Corzine Lab, University of California, Santa Cruz

Sheet: /CAN Transciever1/

File: CANTransciever.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 7/21



Prepared by Thomas Liao

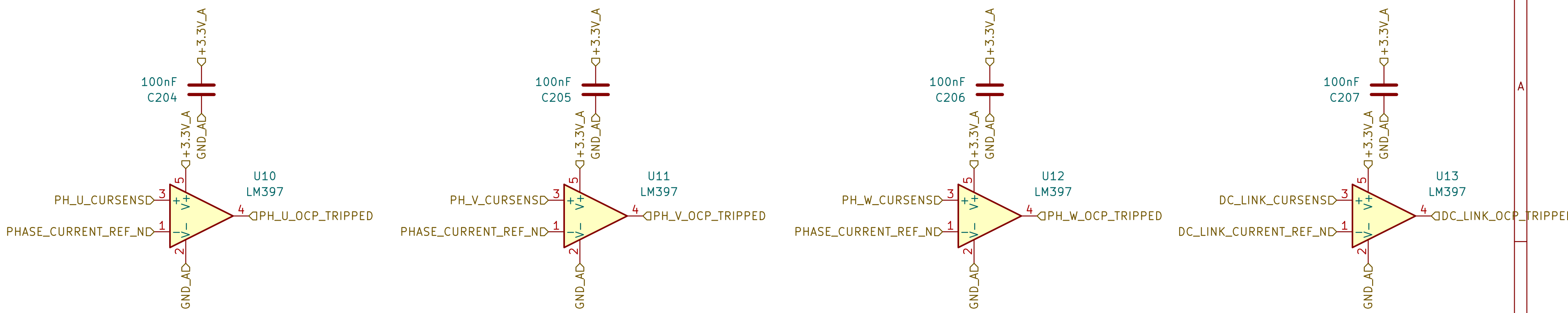
Corzine Lab, University of California, Santa Cruz

Sheet: /Control/

File: Control.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 8/21

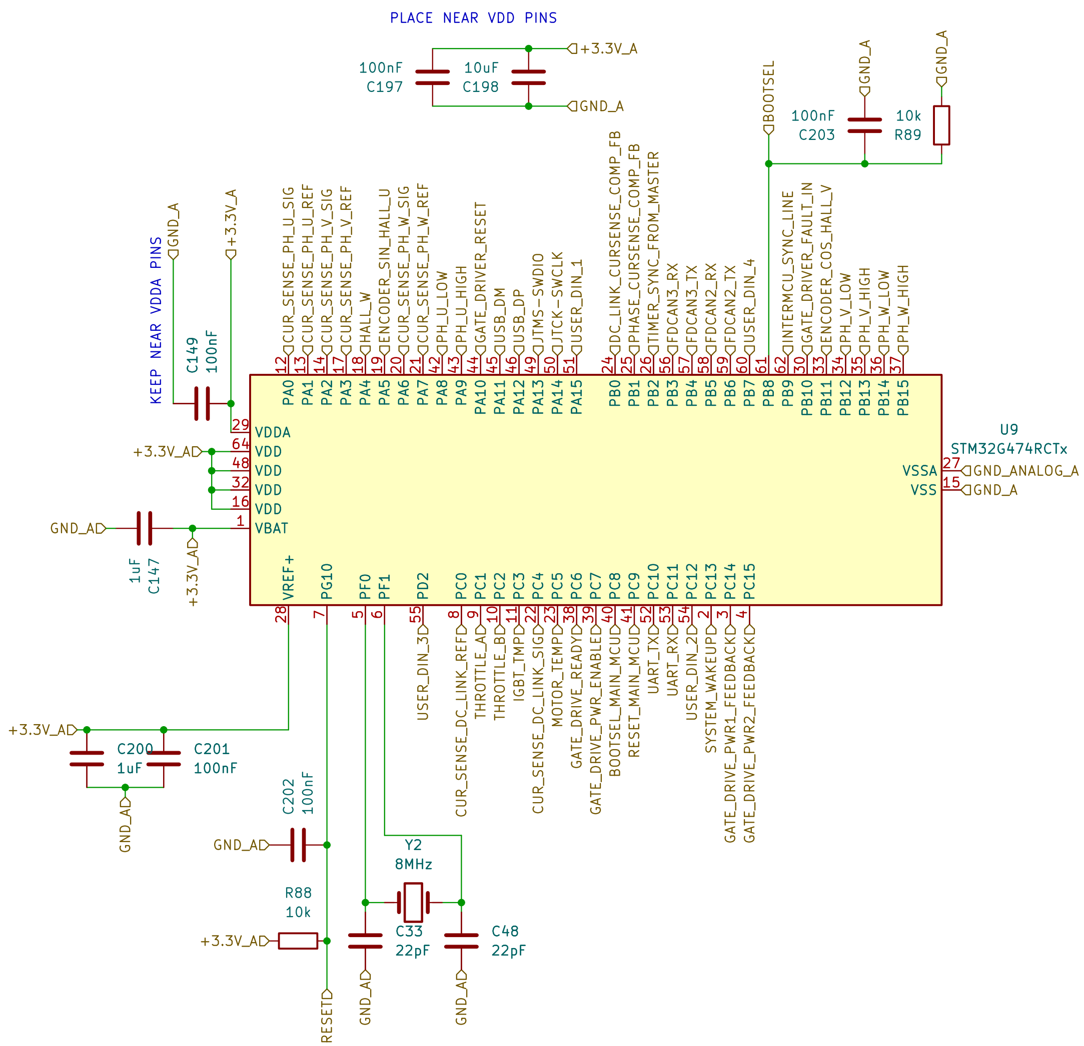


Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Hardware Overcurrent Detection/
File: HWOCP.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 10/21



Prepared by Thomas Liao

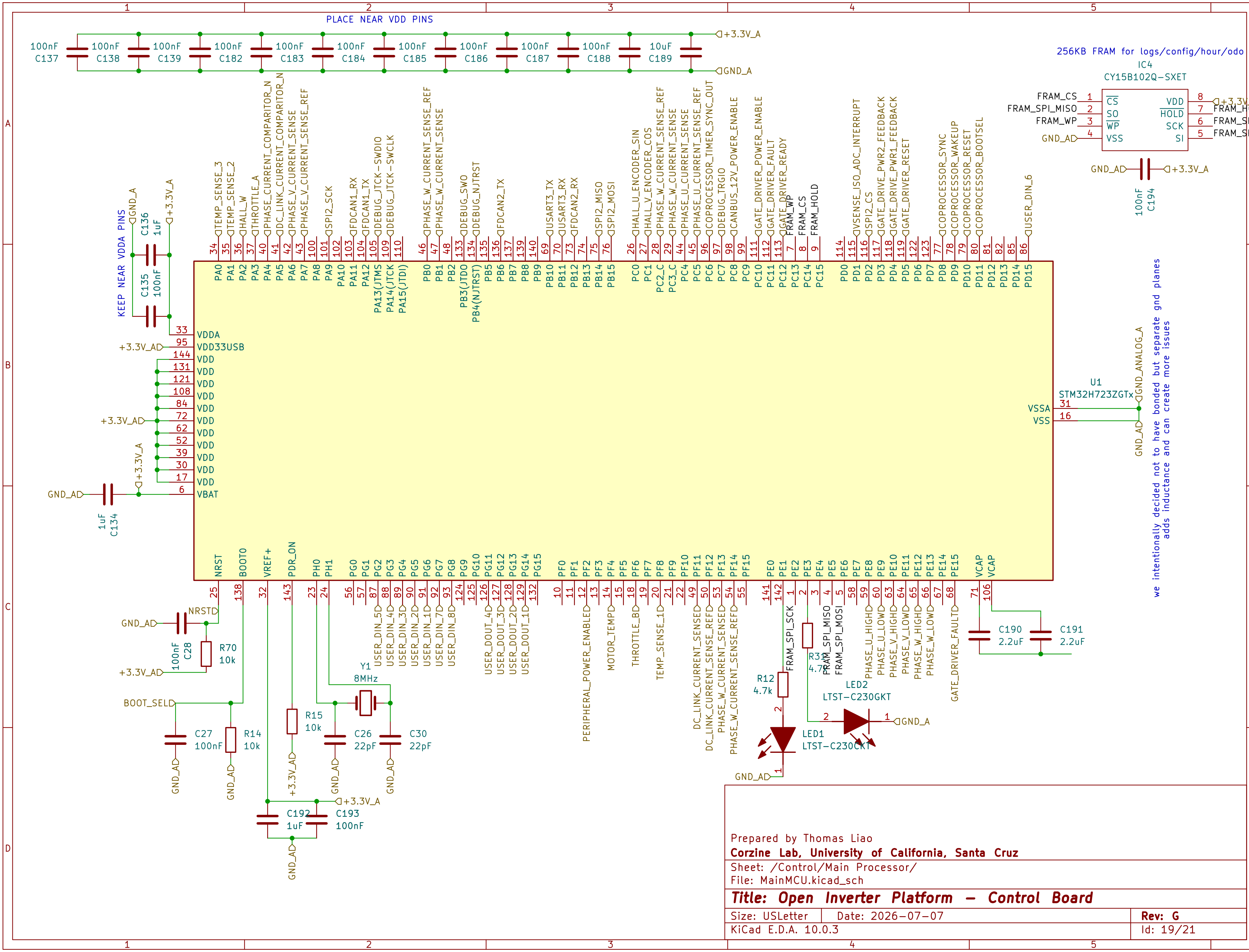
Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Safety Coprocessor/

File: SafetyCoprocessor.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 16/21



Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

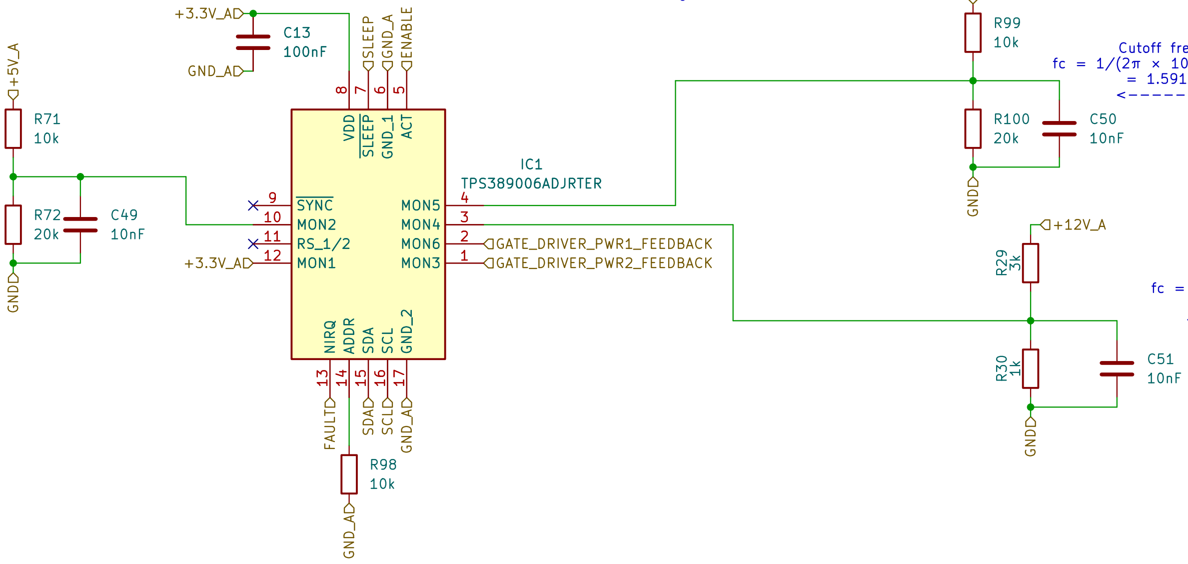
Sheet: /Control/Main Processor/

File: MainMCU.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 19/21

This chip is a pre-certified up to (SIL3) power rail monitor
it is able to assert a fault and disable gate drivers via reset path
can be used to detect brownouts and kill power prior to undefined behavior
is able to monitor rails to within 20mV of targets.



Cutoff frequency:
fc = 1/(2π × 10,000 Ω × 10 nF)
= 1.5915 kHz

Cutoff frequency:
fc = 1/(2π × 10,000 Ω × 10 nF)
= 1.5915 kHz

Cutoff frequency:
fc = 1/(2π × 3,000 Ω × 10 nF)
= 5.305 kHz

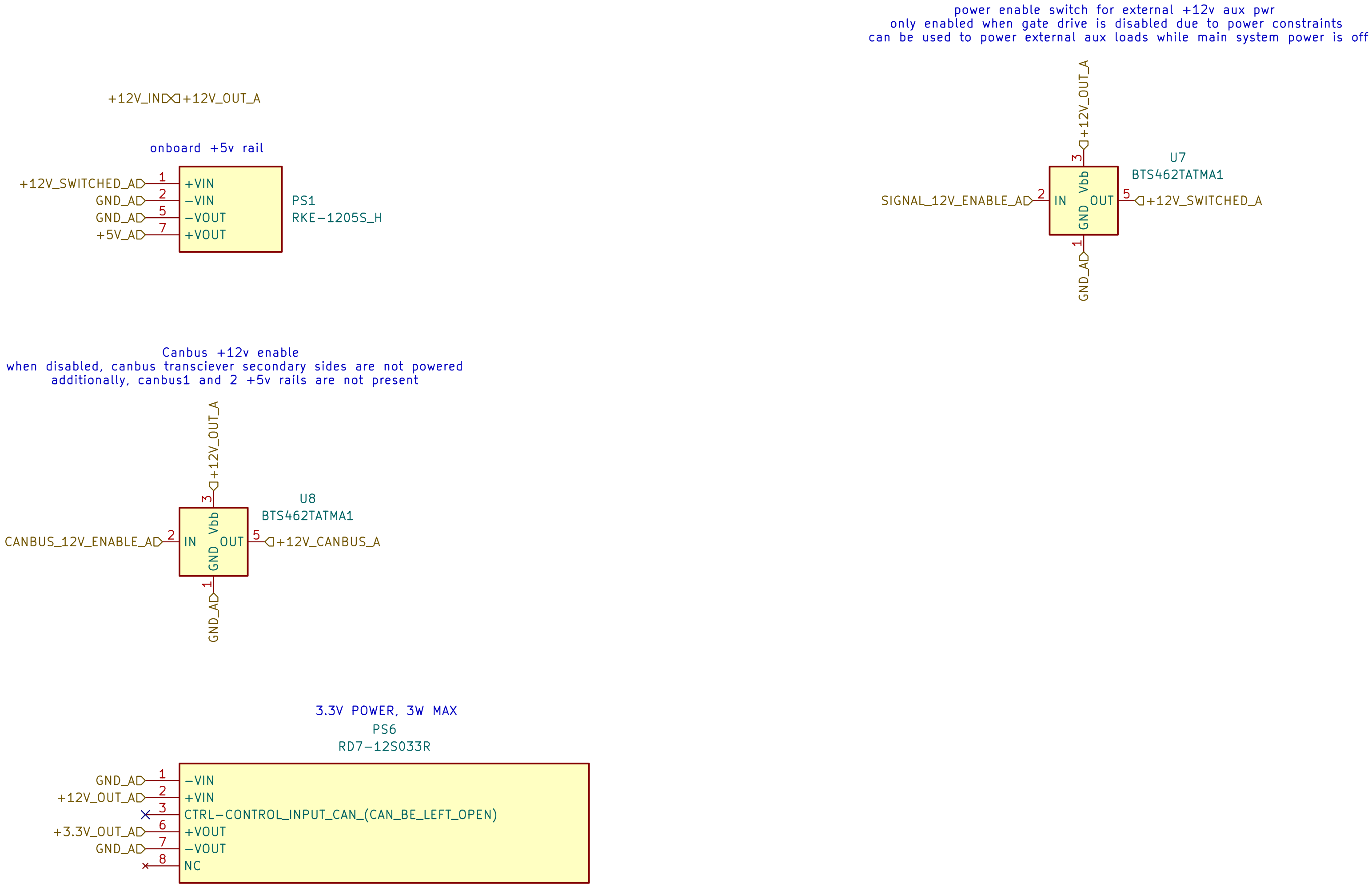
fault here is open collector, on error it pulls down
can be tied in parallel with pwm brkin faults from desat
will auto disable pwm on failure

Sheet: /Control/Power Supply Monitor/
File: PowerSupplyMonitor.kicad_sch

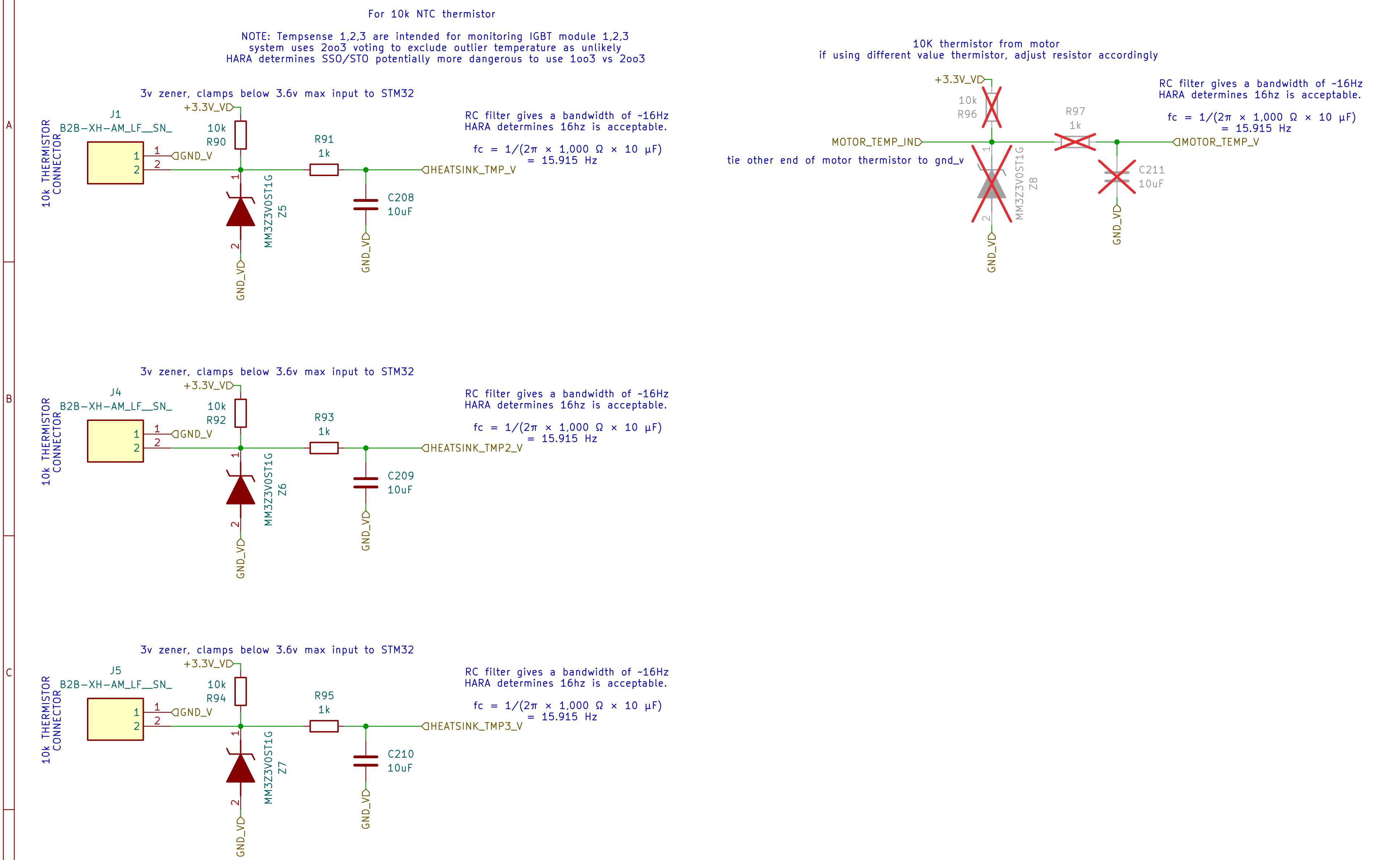
Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.3 | Id: 20/21



Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Power/
File: Power.kicad_sch
Title: Open Inverter Platform - Control Board
Size: USLetter | Date: 2026-07-07 | **Rev: G**
KiCad E.D.A. 10.0.3 | Id: 9/21



Prepared by Thomas Liao

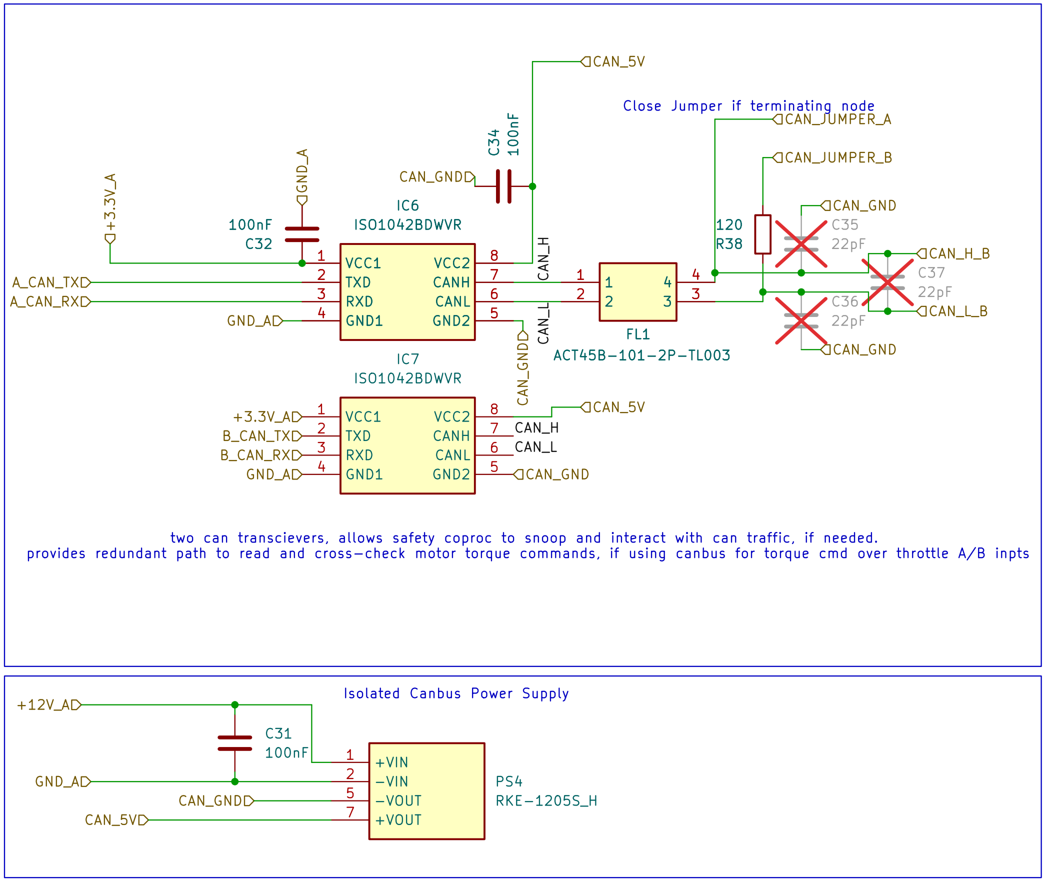
Corzine Lab, University of California, Santa Cruz

Sheet: /TemperatureSensors/

File: TemperatureSensors.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 9/21



Prepared by Thomas Liao

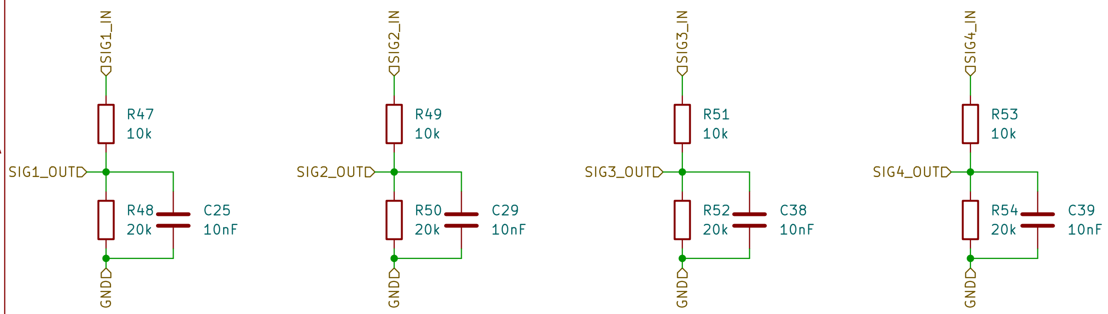
Corzine Lab, University of California, Santa Cruz

Sheet: /CAN Transciever/

File: CANTransciever.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 11/21

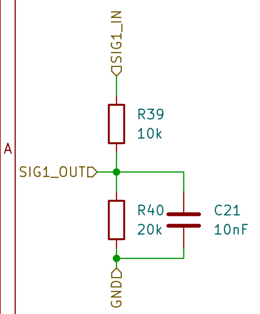


Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network1/
File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 12/21



Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

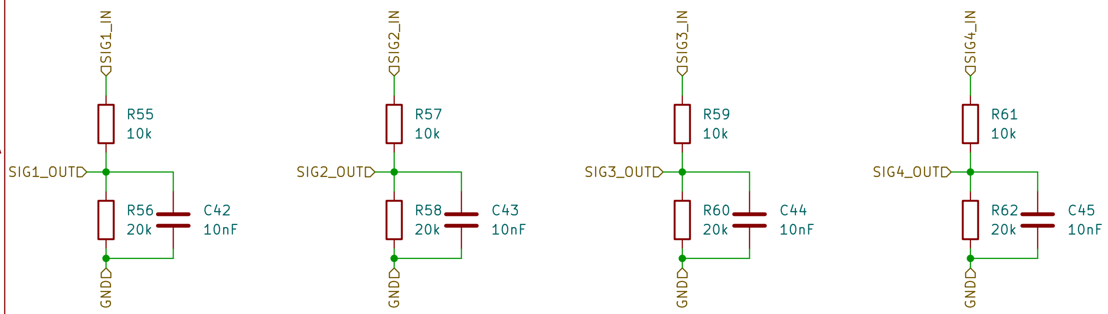
Sheet: /Voltage Divider Network/

File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter | Date: 2026-07-07 | **Rev: G**

KiCad E.D.A. 10.0.3 | Id: 13/21

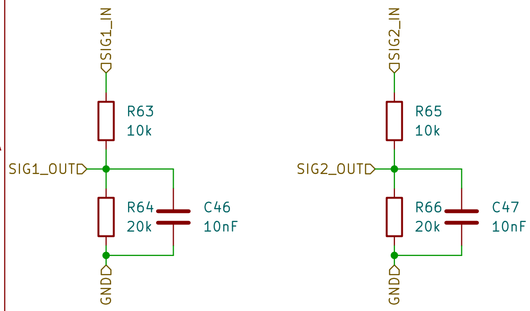


Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network2/
File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

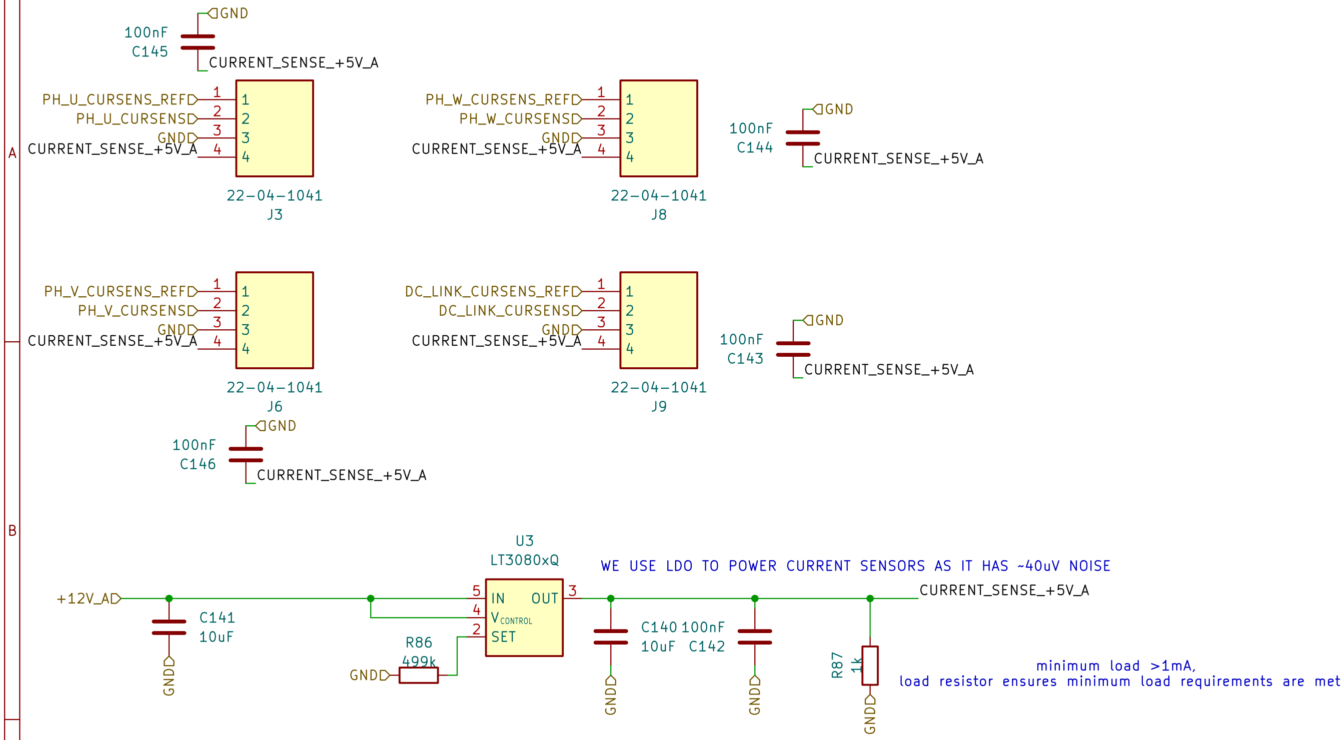
Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 14/21



Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Voltage Divider Network3/
File: VDivNetHalf.kicad_sch

Title: Open Inverter Platform – Control Board

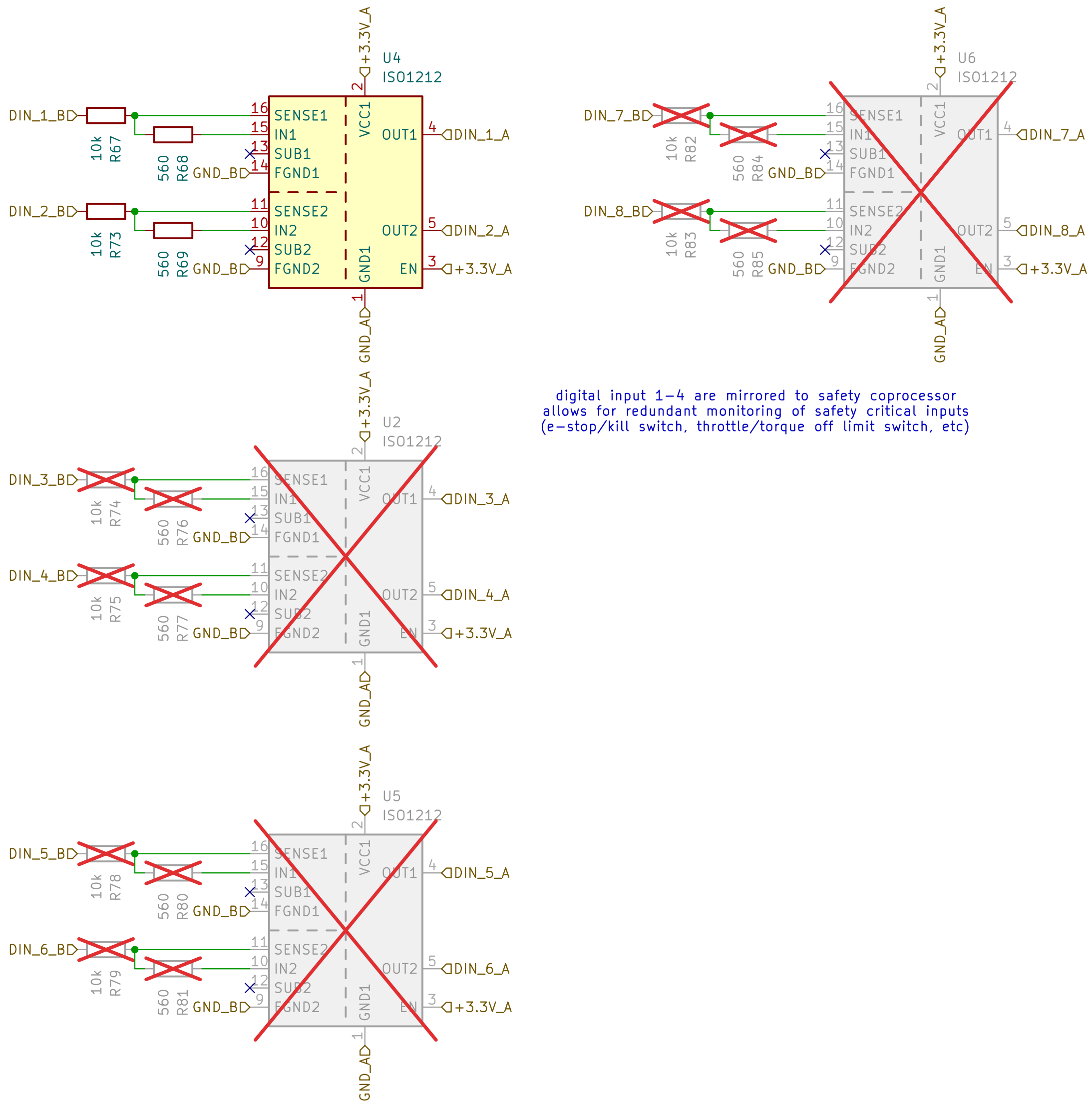
Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 15/21



Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Current Sensors/
File: CurrentSensors.kicad_sch

Title: Open Inverter Platform — Control Board

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 17/21



Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Isolated Digital IO/

File: IsolatedDigitalIO.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: G**
KiCad E.D.A. 10.0.3		Id: 18/21